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Standard Hardware Interface for Sensor Integration in Monitoring Robots

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Abstract

This research presents the development of a standard hardware interface designed to facilitate seamless sensor integration in monitoring robots used for industrial inspection, particularly in the energy sector. The study aims to create a versatile and robust interface that accommodates a wide range of sensors, ensuring compatibility and ease of mounting. The methodology involved a comprehensive analysis of various sensor types, followed by the design and prototyping of a modular interface using 3D modeling and printing. Rigorous testing in industrial settings demonstrated the interface's effectiveness, showcasing quick sensor swaps, reduced downtime, and stable performance under harsh conditions. The results confirm that this standardized hardware interface significantly enhances the adaptability and efficiency of robotic inspection systems, setting a new benchmark for industrial automation and operational excellence in the energy industry.

Introduction

In the market, there exists a wide range of robotic platforms equipped with various sensors, each designed to address specific inspection needs. However, client requirements for autonomous robotic inspections can evolve over time, and it is not always possible for a single robotic solution to meet all these dynamic needs. The diverse array of industrial sensors available can significantly enhance the inspection capabilities of robotic platforms, but integrating these sensors is often a complex process. The presence of different mechanical interfaces can limit and slow down the installation of new devices, making it challenging to adapt to changing requirements efficiently.

To address these challenges, we propose a modular system designed for easy sensor installation on robotic platforms. This modular approach is crucial because, in real industrial use cases, the need often arises to use specific sensors tailored to the client's unique requirements and the assets that need to be inspected. The ability to quickly and easily swap between different sensors is essential for maintaining flexibility and adaptability in various inspection scenarios.

Our modular system has been meticulously designed to overcome the limitations associated with traditional sensor integration methods. It features standardized mechanical interfaces and plug-and-play connectivity, allowing for seamless installation and removal of sensors. This design ensures that robotic

platforms can be quickly reconfigured to meet the specific needs of different inspection tasks, without the need for extensive re-engineering or downtime.

The effectiveness of this modular system has been demonstrated through extensive testing in real-world scenarios. These tests have shown that the system can significantly reduce the time and effort required to integrate new sensors, thereby enhancing the overall efficiency and versatility of robotic inspection platforms. Furthermore, this system is successfully implemented in the currently ongoing European project CARES (Climate-Aware Robotics for Energy Systems), financed by European regional development plan fund ERDF program [1], which focuses on methane leaks detection and remote monitoring in order to meet the European emission regulations.

By adopting this modular approach, we aim to set a new standard for sensor integration in robotic platforms, making it easier for industries to adapt to changing inspection requirements and leverage the full potential of available sensor technologies. This not only improves the efficiency and effectiveness of robotic inspections but also paves the way for more innovative and responsive solutions in the field of autonomous robotics.

Literature Review

In the industry, there are already numerous systems available for hardware and sensor installations, each designed to address specific needs and applications. One of the most well-known standards is the DIN rail [2], characterized by its distinctive "C" or "G" shape profile. DIN rails are primarily used for electrical and electronic equipment, providing a robust and standardized mounting solution for control systems, power supplies, and other industrial components. This standard has been widely adopted due to its reliability and ease of use in industrial settings.

Another prominent example is the Picatinny rail [3], a standardized American accessory rail primarily used in military applications. The Picatinny rail system is designed for mounting optics, tactical lights, lasers, and other accessories on firearms. Its modular design allows for quick and easy attachment and detachment of various components, making it highly versatile in combat scenarios.

The Universal rail system [4] is another innovative solution that employs multiple parallel rails with channels for modular accessory attachment. This system offers exceptional flexibility and expansion options, allowing users to customize their setups according to specific requirements. The Universal rail is particularly popular in applications where adaptability and scalability are crucial, such as in custom-built machinery and specialized equipment.

While these existing systems have their merits, our work differentiates itself specifically through the lateral locking mechanism and the absence of fasteners to lock the instrumentations. We designed and tested this system specifically for robotic industrial applications, with no dual-use case scenarios. This focus ensures that the system is optimized for the unique challenges and requirements of industrial robotics, providing a tailored solution that enhances efficiency and reliability.

Some robotic platforms already come with their own mounting points and mechanical interfaces. For example, the Boston Dynamics Spot quadruped robot is characterized by aluminum rails with T-slot nuts [5], which allow for the attachment of various sensors and tools. Similarly, the Unitree Go2 robot comes with an accessory, produced by third party company, called the "Unitree Go2 Modular Mounting Kit," which enables the addition of Picatinny rail adapters for mounting different accessories. However, each of these robots has its own proprietary interface, leading to a lack of standardization across the market. This fragmentation can make it difficult for users to integrate new sensors and hardware across different robotic platforms, limiting flexibility and interoperability.

To address this challenge, we propose a simple yet effective mechanical system designed to adapt easily to various robotic platforms. Our system can be quickly prototyped in-house, reducing the need for specialized manufacturing processes and lowering costs. The key features of our system include:

- **Lateral Locking Mechanism:** This innovative design allows for secure and quick attachment of sensors and hardware without the need for traditional fasteners. The lateral locking mechanism ensures stability and reliability, even in demanding industrial environments.
- **Standardized Interface:** Our system is designed to be compatible with a wide range of robotic platforms, providing a standardized interface that simplifies the integration process. This standardization promotes interoperability and makes it easier for users to swap sensors and hardware as needed.
- **Ease of Prototyping:** The system's design allows for in-house prototyping, enabling rapid development and iteration. This flexibility is crucial for industrial applications where custom solutions are often required to meet specific inspection needs.
- **No Dual-Use Scenarios:** By focusing solely on robotic industrial applications, our system is optimized for the unique challenges and requirements of this sector. This specialization ensures that the system is highly effective and reliable in industrial settings.

Through extensive testing and real-world implementation, we have demonstrated the effectiveness of our mechanical system in enhancing the versatility and adaptability of robotic platforms. This work represents a significant step forward in the field of industrial robotics, providing a standardized and efficient solution for sensor and hardware integration.

Methodology

The standard hardware interface was meticulously designed to be modular and flexible, capable of adapting to a wide range of instruments. This versatility allows the system to accommodate everything from small, lightweight devices like RealSense cameras and various electronic components to large, heavy-duty industrial sensors such as Optical Gas Imaging (OGI) sensors. The proposed system is built around an octagonal profile linear rail (Fig.1), which serves as the backbone for sensor attachment and support. This design choice provides several key advantages:

Octagonal Profile Linear Rail: The octagonal shape of the rail offers multiple flat surfaces and edges, which facilitate secure and stable mounting of sensors from various angles. This profile ensures that sensors can be supported effectively on all axes, with the exception of the longitudinal direction along the rail. The octagonal design also provides a high degree of structural integrity, making it suitable for both lightweight and heavy-duty applications.

Custom Slider: The system includes a custom-designed slider that moves along the linear rail, allowing for precise positioning of sensors. The slider is engineered to fit snugly within the octagonal profile, ensuring minimal play and maximum stability. This custom slider is a critical component that enables the system's flexibility and adaptability.

The locking mechanism is a crucial aspect of the system, designed to ensure secure and repeatable sensor installation. It consists of an eccentric lever that clips onto specific indentations along the rail. This design allows instruments to be mounted at various discrete positions along the rail, providing the necessary flexibility for different inspection tasks. The locking mechanism offers several benefits:

Discrete Positioning: The indentations along the rail enable sensors to be positioned at precise, repeatable intervals. This feature is particularly useful in industrial applications where consistent and accurate sensor placement is essential for reliable inspections.

Good Installation Repeatability: The eccentric lever design ensures that sensors can be installed and removed repeatedly without compromising the system's integrity or accuracy. This repeatability is vital for maintaining consistent inspection results over time.

To address potential issues with sensor vibrations along the longitudinal axis of the rail, the locking mechanism's tolerances need to be quite tight. This precision helps to minimize unwanted vibrations, ensuring that sensors remain stable and accurate during operation. Alternatively, the locking lever could be

made of a softer material, such as rubber. A rubber lever would compress slightly when engaged, providing extra friction and further enhancing the fixture's stability. This material choice could be particularly beneficial in applications where sensors are subject to significant vibrations or shocks.

Additionally, the system's design considers the ease of use and maintenance. The eccentric lever is designed to be operated quickly and effortlessly, allowing for rapid sensor changes and adjustments. This user-friendly design is crucial in industrial settings where time is often a critical factor.

The proposed system has undergone rigorous testing to validate its performance and reliability. These tests have demonstrated the system's ability to support a wide range of sensors securely and accurately, even in demanding industrial environments. The modular and flexible nature of the system makes it an ideal solution for various robotic inspection tasks, from routine maintenance to specialized surveys.

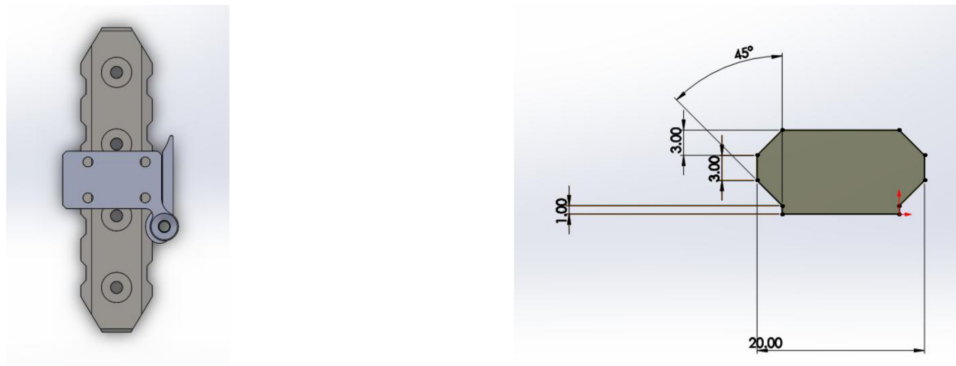


Figure 1—Standard Hardware Interface rail and slider with locking mechanism (Left). Octagonal rail profile (Right).

This design has been implemented in anymal d robotic platform, produced by Anybotics. Specifically, we located several fixture points for the rail installation. The main one is found on the robot body close to the on-board pc ventilation system, additional rails are present also on the robot mobile pan-tilt system, used to accommodate small and lightweight sensors, the additional degrees of freedom of the pan-tilt allow to point the sensor to a specific target without moving the robot body. Finally extra rails have been added to the lidar shield (left side of the image) to provide an extra support for heavy sensors

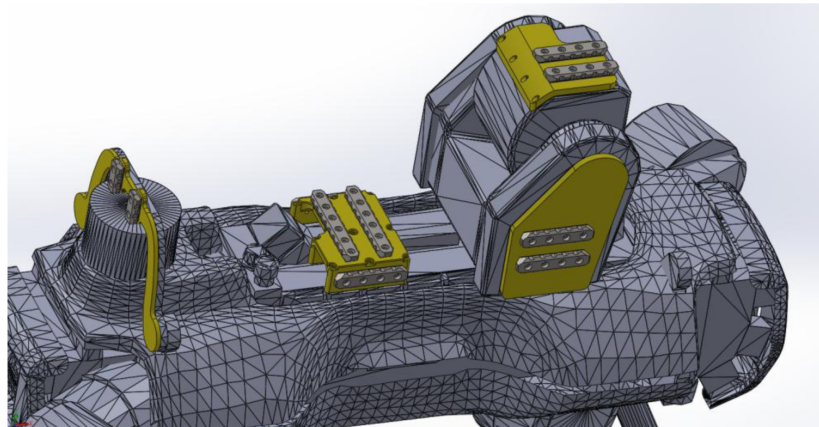


Figure 2—Standard Hardware Interface designed to be installed on Anymal d.

For each specific sensor a bracket has to be designed in order to be installed on the hardware interface. In our work we focused on a wide range of sensors, we then proceeded to the prototyping and testing of their functionalities.

Prototyping and Testing

The rails have been 3d printed with fdm technology, specifically the parts have been prototyped with nylon carbon fiber reinforced (CF-PA6) filament, a high strength material that is characterized by lightweight and high temperature resistance properties.

In our work the following sensors and instruments have been considered:

- Fluke SV600, consisting of an array of ultrasonic microphones and a rgb camera, it's used primarily for leak detection. This sensor counterpart, produced by Sorama, has already been researched and integrated by anybotics [6], we investigated a different installation point instead.
- Flir Grasshopper USB3, a high frame rate camera that can be used for detecting fast transients and motion of machinery.
- Falcon G2, a TDLAS methane sensor specialized for leak detection and fugitive emissions.
- Jetson Orin, a compact pc with high computational power for on edge AI applications.
- POE injector, used for powering the Fluke SV600 and similar devices.
- Sierra Olympia Ventus, an infrared OGI camera for methane detection, this sensor has a specialized detector in the medium infrared range to detect passively the infrared radiation absorbed by methane.

The sensor integration is shown in Fig. 3, the before mentioned instruments can all be installed simultaneously except for Flacon G2, that is supposed to be mounted in place of Flir Grasshopper USB3 (Fig. 4) and the Sierra Olympia Ventus OGI (Fig. 5) that is integrated on the main robot body. The instruments are powered by using the 48v battery connector available on the robot, proper dc-dc converters may be needed for specific sensors. The communication instead is performed through the ethernet port of the robot router; this allows access to the sensor data with a local wireless network. If the ethernet network is not available communication can be integrated with the onboard robot USB type c port. Sierra Olympia Ventus OGI (Fig. 5) has been integrated with extra care, specifically vibration dampers have been introduced to reduce the vertical accelerations of the delicate sensor's electronics during the robot locomotion, a dedicated dc-dc 48v-12v converter has been integrated in the package making one simple compact solution. Finally, communication is achieved with the sensor ethernet port.

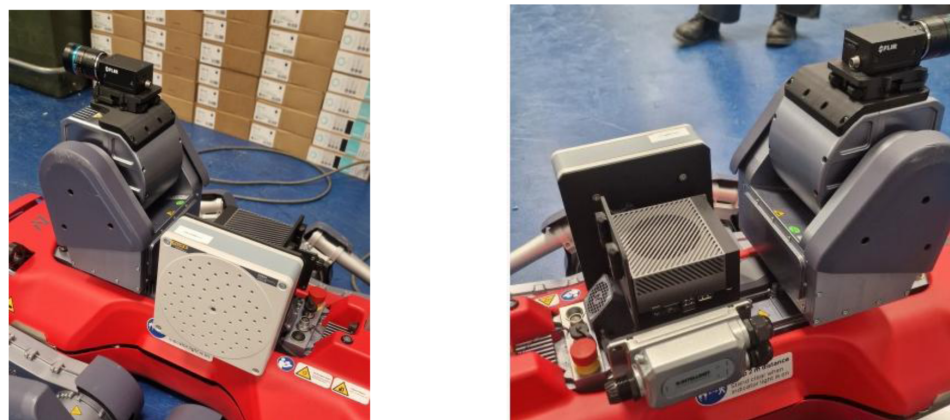


Figure 3—Sensors installed through the standard hardware interface.



Figure 4—Falcon G2 installed through the standard hardware interface.



Figure 5—Ventus Sierra OGI camera installed through the standard hardware interface.

Results and Observations

All the instruments integration described in the previous section have been tested with the robot operating in real Baker Hughes industrial facilities. We focused specifically to the instruments stability fixed with the standard hardware interface, to the movements of the robot and to the battery duration. The sensors appeared stable and well secured in place during all the testing period of time, the robot produces a significant amount of vibrations during its locomotion, this fact could loosen screws and fastener, the proposed locking mechanism design instead is not susceptible to these, making this solution ideal for a fast and easy sensor replacement. Furthermore, from the conducted test, the robotic platform appeared to be quite robust and was able to carry all the instruments with minor effects on the robot locomotion itself; the maximum payload weight, declared by the manufacturers, is up to 15kg without the pan-tilt system, the robot movements in fact appeared to be quite repeatable, adapting to varying loads without the risk of the robot falling. However extra attention should be paid in the presence of difficult terrains (e.g. staircases) or narrow passages, each situation must be assessed specifically and extra care and / or protection should be disposed to avoid potential damage to expensive equipments. The battery duration appeared to be able to provide all the necessary power to the instruments, current spikes were not an issue with the 48v unregulated power port. The robotic platform comes also with two additional regulated power ports 12v and 5v, however these ones are limited in current and susceptible to current spikes, inductive loads should be avoided in this case and their use should be assessed for each specific device. The duration of the battery appeared to be sufficient for regular inspection operations and for powering all the devices at the same time, the majority of sensors as a matter of fact are characterized by a low power consumption.

Next developments and future works

For the next developments of the standard hardware interface, several key advancements will be pursued to enhance performance, durability, and user experience. One of the primary areas of focus will be the testing of new materials, specifically titanium and aluminum alloys. These materials are chosen for their exceptional

strength-to-weight ratios, making them ideal for applications where both lightweight and robust construction are critical. Titanium alloys, known for their high strength, corrosion resistance, and biocompatibility, will be evaluated for their suitability in demanding environments. Aluminum alloys, on the other hand, offer a good balance of strength, lightweight, and cost-effectiveness, making them a viable option for various components.

In parallel, a comprehensive redesign of the guide and the locking mechanism will be undertaken. The current design will be scrutinized for potential improvements, with a particular emphasis on enhancing ease of use and reliability. One innovative solution being considered is the adoption of a ratchet-like mechanism. This mechanism would allow for the automatic locking of sensors in place, ensuring precise alignment and secure attachment without the need for manual adjustments. Such a feature would not only streamline the installation process but also reduce the likelihood of human error, thereby improving overall system performance.

Additionally, the reliability and robustness of the redesigned system will be rigorously assessed in real-world conditions. This will involve conducting extensive testing in clients' facilities during routine maintenance inspections. By subjecting the hardware interface to actual operational environments, we can gather valuable data on its performance under various conditions. This feedback will be crucial in identifying any potential weaknesses and making necessary adjustments to ensure the system meets the highest standards of reliability and durability.

Furthermore, the testing phase will include stress tests, environmental simulations, and endurance trials to evaluate the system's resilience against wear and tear, temperature fluctuations, and other environmental factors. The results of these tests will inform further refinements, ensuring that the final product is not only technologically advanced but also practical and reliable for long-term use.

In summary, the next phase of development for the standard hardware interface will involve material innovation, mechanical redesign, and thorough field testing. These efforts aim to create a more efficient, durable, and user-friendly system that can withstand the rigors of real-world applications and meet the evolving needs of our clients.

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